Lesson 11 Counting

In Lessons 11 to 13, we explore another type of mathematical reasoning: **counting arguments** and **combinatorial proofs**. We begin with some basic concepts in counting, the first is the **multiplication principle**, which enumerates (**ordered**) **lists** of k elements, $(a_1, a_2, \ldots, a_k)$. Suppose there are n_j choices for the jth element a_j of a k-element list. The **multiplication principle** (**MP**) says that the number of such lists is the product $n_1 n_2 \cdots n_k$. Lists are *not* sets:

- Order matters for lists but not sets: $(1,2,3) \neq (2,1,3)$ but $\{1,2,3\} = \{2,1,3\}$.
- The elements of lists may repeat but not for sets: $(1,1,) \neq (1)$ but $\{1,1\} = \{1\}$.

We will give a first application of the multiplication principle.

Definition The ***cardinality*** $|A|$ of a finite set A is the number of elements in A.

Proposition 1 If A and B are finite sets, then $|A \times B| = |A| \times |B|$.

Proof The Cartesian product $A \times B$ is the set of ordered pairs, or 2-element lists, (a, b). There are $|A|$ choices for the first element a and $|B|$ choices for the second element b. By the multiplication principle, there are $|A| \times |B|$ pairs.

Consider a k-element list where each element is one of the same n possibilities.

- If repetitions are allowed, then $n_1 = \cdots = n_k = n$, and the number of lists is
$$n \times n \times \cdots \times n = \boldsymbol{n^k}$$
- If repetitions are *not* allowed, then the number of lists is
$$n(n-1)\cdots\big(n-(k-1)\big) = n(n-1)\cdots(n-k+1) = P(n,k)$$
The number $\boldsymbol{P(n,k)}$ is the number of **permutations** (or **arrangement**) of k objects from n objects. If $n < k$, then one of factor is 0, and $P\,(n,k) = 0$, which corresponds nicely to the fact that it is not possible to make a list.

The product in $P(n,k)$ may remind us of **factorial**: $n! = n(n-1)\cdots(3)(2)(1)$, for a positive integer n, with $0! = 1$. For $n \geq k$, we have:

$$\begin{aligned} P(n,k) &= n(n-1)\cdots(n-k+1) \\ &= \frac{n(n-1)\cdots(n-k+1)(n-k)(n-k-1)\cdots(2)(1)}{(n-k)(n-k-1)\cdots(2)(1)} = \frac{n!}{(n-k)!} \end{aligned}$$

Note that $P(n,n) = n!$, there are $n!$ ways to arrange n objects in a line. Also,
$$P(n,1) = \frac{n!}{(n-1)!} = n$$

Suppose the password for a system is 5 characters long such that the first three characters are lower case letters and the last two characters are digits.

a) The number of possible passwords is:
$$26 \times 26 \times 26 \times 10 \times 10 = 26^3 \times 10^2$$
b) The number of passwords with all distinct (different) characters is

$$26 \times 25 \times 24 \times 10 \times 9$$

c) How many passwords end with a prime digit $(2, 3, 5, 7)$?

$$26^3 \times 10 \times 4$$

d) How many passwords have no vowels? There are 5 vowels: a, e, i, o, u.

$$21^3 \times 10^2$$

e) How many passwords have at least one vowel?
At least one means "not zero", subtract from all passwords those with zeros:

$$26^3 \times 10^2 - 21^3 \times 10^2$$

How many ways can a family of 2 parents and 4 kids sit in a row if the kids sit together? We count in two steps. First there are 3! ways to arrange 3 groups: two groups of a parent and a group of 4 kids. Next there are 4! ways to arrange the kids. So, the answer is 3! 4!

We move on to another basic counting problem, that of **combinations**:

How many subsets of size k does an n-element set have?

We denote this number $\binom{n}{k}$ and call it ***n* choose *k***; it is the number of ways to choose k objects out of n objects, *unordered*. We will find a formula for $\binom{n}{k}$ by connecting $\binom{n}{k}$ to $P(n,k)$ using a **combinatorial argument**. To make ordered lists of length k, we first select a subset of k elements, there are $\binom{n}{k}$ ways to do this. We then arrange the k chosen elements, and there are $k!$ ways to do this, so:

$$P(n,k) = \binom{n}{k} k!$$

$$\Rightarrow \qquad \binom{n}{k} = \frac{P(n,k)}{k!} = \frac{n!}{k!\,(n-k)!}$$

This formula makes sense for the special cases $k = 1$ and $k = n$: there are n ways to choose 1 object and one way to choose n objects from n objects:

$$\binom{n}{1} = \frac{n!}{1!\,(n-1)!} = n, \qquad \binom{n}{n} = \frac{n!}{n!\,(n-n)!} = 1$$

We give two numerical examples.

1) Let $A = \{1, 2, 3, 4, 5\}$. How many subsets of 3 elements are there?

$$\binom{5}{3} = \frac{5!}{2!\,3!} = 10$$

2) A password for a system has 6 digits. How many passwords have exactly two 5's? We first choose 2 out of 6 places to put the two 5's, and there are $\binom{6}{2}$ ways of doing this. Then in each of the 4 remaining places we put in one of the nine digits other than 5. So the answer is,

$$\binom{6}{2} \times 9 \times 9 \times 9 \times 9 = \binom{6}{2} 9^4$$